Living Village – Assignment 4: Production Plan & MVP 2.0 (Alpha)

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Playable alpha: https://ishashah2303.github.io/LivingVillage/ (the LLM planner runs on Render and can take a minute to wake up; until then the same rules run in the browser) · **Source:** github.com/ishashah2303/LivingVillage

1. Production schedule (Weeks 10–15)

Week	Engineering	Assets	Integration / milestone
10 · Nov 2–8	**Scale the cast** from 8 to 32–64 villagers: generated villagers, staggered behavior-tree ticks with a distance-based think rate, spatial hash for perception, instanced rendering. Replace the greedy evacuation assignment with **min-cost flow** (capacity-constrained, optimal).	AI-generated low-poly villager (Meshy / Tripo), rigged with Mixamo idle / walk / run / work clips.	Crowd-size switch in the stress test; benchmark rerun at 8 / 32 / 64 villagers.
11 · Nov 9–15	**More emergency types** through one hazard abstraction: fire, flood at the pond, storm (blackout + falling trees), sickness. Generalize the command grammar (`SHELTER_IN_PLACE`, `FETCH(item)`) and the safety gate. Danger-area costs in the navmesh (Detour query filter) so routes avoid smoke, not just obstacles.	Flood water and rain shaders, AI-generated storm skybox, hazard icons.	Fire Season becomes a "Season" mode with mixed emergencies.
12 · Nov 16–22	**Hierarchical LLM planning**: cluster villagers, let the rules / min-cost flow assign individuals and let the LLM decide group-level and social choices, so the prompt grows with groups, not people. **Model cascade**: Haiku first, escalate to Sonnet when grounding drops commands or the crowd is large; prompt caching for the fixed system prompt. pgvector memory retrieval into the prompt.	Animation blending on the rigged characters; ambient and UI sound (procedural fire crackle, bell).	**Beta (feature complete). Playtest with 5 classmates; log time-to-safety and goal switches.**
13 · Nov 23–29 Thanksgiving	Evaluation suite: per-backbone and per-crowd-size metrics (time to safety, goal switches/min, stuck recoveries, commands accepted vs. rejected, cost per game). Boids in a Web Worker. Balance pass from playtest data.	Menu art and logo, end-screen illustrations.	Short believability survey (playtesters rate villager behavior).
14 · Nov 30–Dec 6	**Feature freeze.** Low-end GPU settings (shadow quality, DPR cap), keyboard and color accessibility, final benchmark run with all backbones.	Demo video capture and edit; final screenshots.	**Release candidate** deployed **(GitHub Pages + Render).**
15 · Dec 7–13	**Buffer week – no new features.** Bug fixing from the RC, documentation (README, architecture, benchmark write-up), polish, final report and submission (due ~Dec 11).		

2. MVP 2.0: the full loop

Start state: a menu over the live village picks the mode (*Fire Season* challenge or *Sandbox*) and the planner's LLM backbone (Opus 5.5, Sonnet 5.5, Haiku 5.5 or rules only). **Active state:** in Fire Season the player has 4 minutes until the rains come. A director starts fires faster and faster, unattended fires spread, smoke hurts villagers, and a food shortage hits midway. The player has water buckets (refilling), food baskets and an alarm bell; the villagers' behavior trees, the Detour crowd, the boids and the LLM planner do the rest. The HUD shows time left, how much of the village stands, who is hurt and what is burning. **End state:** the village wins when the rains come (1–3 stars and a score) and loses at 0% integrity or 3 villagers hurt. Both lead to a results screen (time held out, fires put out, planner commands accepted / rejected, mean planner latency, LLM cost), then *Play again* or *Main menu*. Sandbox keeps the Assignment 3 world editor and ends with a session summary. All pillars run inside the game: the behavior tree got a new branch (a hurt guard stops fighting and flees), navmesh obstacles change with every fire, the flock scatters from fires, and the planner coordinates evacuations.

3. Release notes (since Assignment 3)

New features

- Game loop: start menu, Fire Season challenge, win / loss / session end screens, Play again and Main menu (each game gets a fresh simulation).
- HUD (timer, village integrity, hurt, burning, pause banner); player tools with charges and cooldown; keys 1–3, `B`, `Space`.
- Game director with escalating waves; fire spread; health, smoke damage and starvation; injured villagers leave the map.
- Planner backbone chosen per game in the menu; every plan reports its tokens and list-price cost (planner panel and end screen).
- Backbone benchmark (`server/bench`) over scenarios exported from real simulation runs.
- Tests: 31 simulation tests (+11, including full games played headless: an idle player must lose, a simple bot must win) and 18 server tests (+2).

Bug fixes

- Haiku plans were cut off at `max_tokens` 4096 on large crowds and came back empty; raised to 16k (found by the benchmark).
- Server grounding did not enforce the 3-villagers-per-spot limit; it now drops the extra command.
- A planner reply arriving after a restart could touch a freed navmesh; the WASM navmesh is now disposed on restart and late replies are ignored.
- The "Stale plan" demo left the simulated latency at 4 s for the rest of the session.
- The top bar overflowed at phone width.

Known issues

- LLM latency has long tails (one Opus call took 92 s, one Haiku call 64 s); after 30 s the rules answer instead, so the village never waits, but that plan is lost.
- A Fire Season game sends about 15–20 plans; the hosted planner's spend cap (20 Claude plans per visitor per hour) makes a second game in the same hour fall back to the rules.
- Evacuation coverage drops to ~65% at 32 villagers for every planner: 11 safe spots × capacity 3 is not enough (Week 10).
- Villagers asleep indoors ignore a fire next to their house.
- Balance is tuned with headless bot games only; no human playtest yet (Week 12).
- Characters are still procedural, there is no audio, and the free Render tier sleeps after 15 idle minutes.

4. Response to feedback: other LLM backbones and scaling

Same prompt, schema and grounding for every model, effort `low`, 7 scenarios exported from real runs (5 with 8 villagers, 2 synthetic crowds of 24 and 32), 2 runs each; 56 calls cost about $0.82 in total. *Coverage* = share of villagers within 14 m of a fire given a valid evacuation; *validity* = share of the model's commands that pass grounding.

Planner	Median latency (max)	Cost / plan, 8 → 24–32 villagers	Valid	Coverage	Est. cost / game*
Claude Opus 5.5	9.7 s (92 s)	$0.023 → $0.078	100%	94%	~$0.40
Claude Sonnet 5.5	4.2 s (15 s)	$0.011 → $0.032	97%	91%	~$0.19
Claude Haiku 5.5	11.1 s (64 s)	$0.001 → $0.003	100%	94%	~$0.02
Rule-based	<1 ms	$0	100%	94%	$0

* about 17 plans in an 8-villager Fire Season game. Prices: Opus $4/$20, Sonnet $2/$10, Haiku $0.10/$0.50 per million input/output tokens.

- **Backbones.** On our task Sonnet 5.5 is the best default: fastest (4.2 s median), half of Opus's cost, similar coverage; its 3% invalid commands (spots too close to a fire) are all caught by grounding. Haiku is ~20× cheaper than Opus with equal coverage but thinks long even at low effort (2.5k output tokens on average), so it is *not* faster. The menu now defaults to Sonnet, and Week 12 adds a Haiku-first cascade.
- **Scaling.** The prompt grows linearly with the cast (2.9k → 8.9k input tokens from 8 to 32 villagers; Sonnet and Opus latency ×2–2.6), and coverage is capped by spot capacity, not by the model. Planned algorithms: min-cost flow for capacity-constrained assignment and flow fields for mass evacuation (W10); hierarchical planning so the LLM plans per group (W12); BT time-slicing, spatial hashing and instancing for characters and objects (W10); one hazard abstraction with per-hazard navmesh area costs for new emergency types (W11).